LEARNING MODULE

19

Descriptive Statistics of Returns

A statistic of returns is a choice of divisor, quantile rule and period. On 2003–2025, TAIEX's days scale to an annual volatility of 18.56%, its calendar years give 22.21%, and the same years cut at other days give anything from 11.89% to 25.04%.

VOLUME IV · Statistical Inference and Econometrics

BUILDS ON LM1 · LM3 · LM4 · LM5 · LM10 · LM14 · LM15 · LM16 · LM17 · LM18

LEADS TO LM20 · LM21 · LM22 · LM23 · LM24 · LM25 · LM26 · LM27 · LM32 · LM33

TIME ≈ 3 h + lab

LEARNING OUTCOMES

Mastery	*After this module you should be able to:*
☐	1 calculate the mean, median, trimmed mean, compound rate and mean log return of a return series, and explain why they differ
☐	2 calculate the standard deviation with divisor T and $T-1$, the mean absolute deviation, the interquartile range and the median absolute deviation, and convert the last two into normal-equivalent volatilities
☐	3 compute sample skewness and excess kurtosis, their adjusted versions and quantile-based measures, and judge which of them hang on a few days
☐	4 compute the sample autocorrelations of returns, absolute returns and squared returns under the book's convention, and read them against the iid band
☐	5 derive the variance of a sum of m days from autocorrelations, estimate the variance ratio from overlapping and non-overlapping sums, and state the overlapping estimator's bias
☐	6 apply the annualization rules for the mean, variance, volatility and Sharpe ratio, state the condition each needs, and explain why TAIEX's annual volatility depends on how it is measured
☐	7 build weekly, monthly and yearly returns from daily data, and compute every statistic of the module in Python with explicit divisors and quantile rules

LEARNING MODULE OVERVIEW

- TAIEX's day has five centres: mean 0.0625%, median 0.0879%, 5% trimmed mean 0.0832%, compound rate 0.055185% and mean log return 0.055170%. The median and the trimmed mean sit above the mean because large falls outnumber large rises.
- The standard deviation divides by T ($s = 1.2084\%$) or by $T - 1$ ($s_{\mathrm{u}} = 1.2085\%$), (19.1). The normal-equivalent robust scales of (19.2) give 0.873% and 0.868%: the typical day moves like a 13.7% annual volatility, against 18.96% from s.
- Skewness and kurtosis, (19.3), hang on a few days. One more −10% day moves TR23's skewness from −0.338 to −0.425 and its excess kurtosis from 5.27 to 5.88, while the quantile measures of (19.4) move by 1.2% and 0.05%.
- Daily log returns barely correlate with their past ($\hat{\rho}_1 = 0.0393$, (19.5)), but absolute log returns correlate between 0.096 and 0.217 at every lag up to 30, far outside the iid band of ±0.0256, (19.6).
- The variance of a sum of m days depends on every autocorrelation up to lag $m - 1$, (19.7). The overlapping variance ratio of (19.8) is 1.208 at one year and 0.619 at five years on TR23, against 0.958 and 0.790 expected for iid days.
- Means of log returns annualize by A exactly; their variances, volatilities and Sharpe ratios by A or $\sqrt{A}$ only for uncorrelated days, (19.9). On simple returns these rules are conventions: TR23 averages 15.38% a year, with volatility 18.96% and Sharpe ratio 0.0517 a day or 0.81 a year, with $r_f = 0$.
- On 2003–2025, TAIEX's annual volatility is 18.56% from days, 19.05% from overlapping one-year sums and 22.21% from calendar years. Cut into one-year blocks from other days, the same 23 years give 11.89% to 25.04%: most of the gap reflects how few years there are.

1 INTRODUCTION

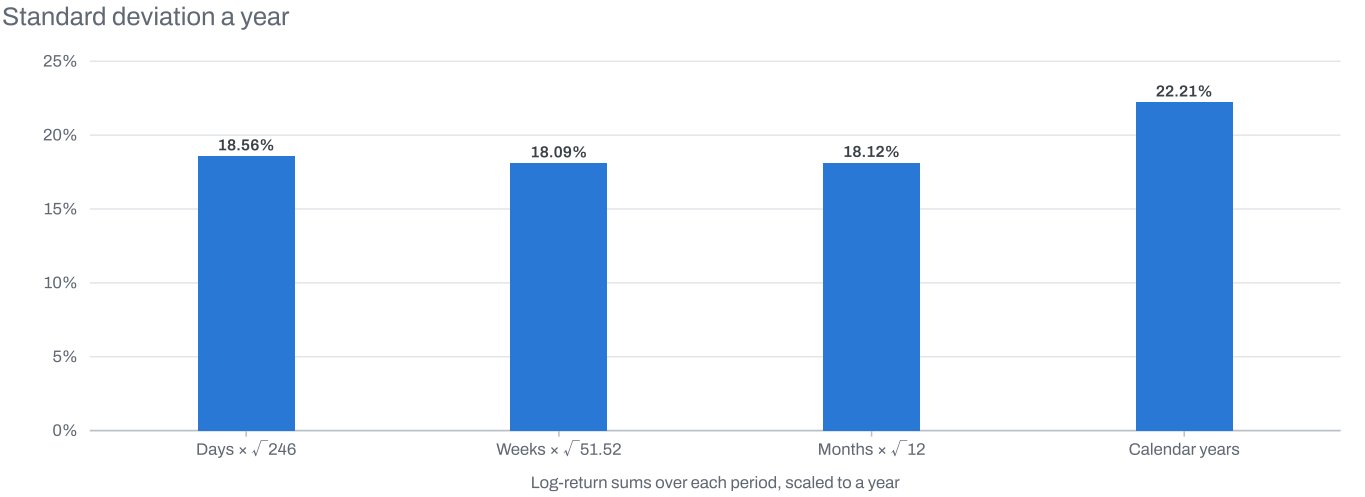
Volume III treated returns as random variables; this module describes them as data. Each statistic of a return series is a recipe: which returns, which divisor, which quantile rule, which period. The case shows how far one annual figure moves with the recipe.

MOTIVATING CASE **How volatile is a year of TAIEX?**

Take the TAIEX total-return index, with dividends reinvested and no costs, from 3 January 2003 to 31 December 2025: 5,664 daily log returns. Their standard deviation is 1.18352% a day (LM17). A year has about 246 sessions, so independent days would give a yearly standard deviation of $\sqrt{246} \times 1.18352\% = 18.56\%$. Weekly and monthly log returns, scaled to a year the same way, give 18.09% and 18.12% (Exhibit 1).

Yet the 23 calendar-year log returns have a standard deviation of 22.21% (LM15). To reach it, the variance of a year's sum would have to be 1.43 times the sum of its daily variances. Do daily variances fail to add up over a year? Or do 23 numbers pin down a standard deviation too loosely to tell?

Exhibit 1: TAIEX's annual standard deviation, measured four ways



Note: standard deviations of 2003–2025 log-return sums (2003 from its first close) over 5,664 days, 1,185 ISO weeks of one to six sessions, 276 months and 23 years. Each divides by the number of periods (LM15's convention). A year is 246 sessions (LM17), 1,185/23 = 51.52 weeks or 12 months.

The sample keeps its 23 Saturday sessions, from 14 October 2006 to 22 December 2018. Each of the 17 six-session weeks holds one, and the other six fall in weeks of four or five sessions. TWSE's history page states that the market stopped trading on Saturday make-up workdays from 2019; that these sessions were such days is an inference. Source: FinMind `TaiwanStockTotalReturnIndex` (data_id TAIEX), retrieved 18 October 2026; dividends reinvested, no costs. Computations: `data/lm19_numbers.py` and `data/lm19_charts.py`.

Sections 2 to 4 describe one day, Sections 5 and 6 how days depend on each other and add up. Section 7 annualizes and resolves the case, and Section 8 turns it all into Python.

NOTE **Notation for this module**

TR23 is the TAIEX total-return index: $T = 5{,}849$ daily returns (a count), 3 January 2003 to 8 October 2026, over 23.7645 years from the first close, so $A = 246.12$. Simple returns are R_t, log returns $r_t = \ln(1 + R_t)$. Sample moments $\bar{R}$, s, $\bar{r}$, s_r divide by T and s_{u} by $T - 1$, (19.1); model moments are μ, σ, g, σ_r. Quantiles are $\hat{F}^{-1}(u) = R_{(\lceil uT \rceil)}$ and $\mathrm{med} := \hat{F}^{-1}(\frac{1}{2})$ (LM14); LM5's $\hat{\mu}$ is $\bar{R}$.

Local: j_0 the returns trimmed at each end; IQR, MAD the interquartile range and the median absolute deviation; $\widehat{\mathrm{Skew}}$, $\widehat{\mathrm{Kurt}}$ the sample skewness and kurtosis, with subscript adj for the adjusted versions; $\mathrm{Skew}_{\mathrm{Q}}$, $\mathrm{Kurt}_{\mathrm{O}}$ the quartile skewness and octile kurtosis; x_t, $\bar{x}$ a series (r_t, $|r_t|$ or r_t^2) and its mean; γ_k, ρ_k and $\hat{\gamma}_k$, $\hat{\rho}_k$ the autocovariance and autocorrelation at lag k, model and sample; $r_t^{(m)} := r_{t-m+1} + \cdots + r_t$ the log return over m days; $\mathrm{VR}(m)$, $\mathrm{VR}_{\mathrm{n}}(m)$ the overlapping and non-overlapping variance ratios, (19.8), and o the returns skipped before the first block; SR a Sharpe ratio with $r_f = 0$, and $\mathrm{SR}(m)$ that of sums of m days. Counts: j a rank, k a lag, m the days in a sum. The model's ratio in (19.7) gets no symbol, so that $\mathrm{VR}(m)$ keeps one meaning.

NOTE **Research hygiene**

Every TAIEX figure describes TR23, unless the text or a caption names a limit regime or 2003–2025 (LM15's and LM17's calendar years). The figures are **descriptive**, without standard errors, which are LM20's subject.

The normal law with the sample's own mean and standard deviation is **in-sample**. The normal-law constants of (19.2), the iid band and expectation, and experiment 1 are **model** properties, and the extra −10% day is **hypothetical**. The limit regimes, and 2003–2025 against TR23, are **stability checks**. Nothing is tested (LM21, LM25).

2 WHERE RETURNS CENTRE

- ☐ calculate the mean, median, trimmed mean, compound rate and mean log return of a return series, and explain why they differ

A centre is one number that stands for a whole series. Returns have several, and in a week of extreme days they disagree.

EXAMPLE 1

The tariff week

TR23's returns on 7–11 April 2025, the week after the United States announced new tariffs (The White House 2025), were −9.6997%, −4.0161%, −5.7865%, +9.2512% and +2.7842%. With a standard deviation of 6.72%, this was the most volatile of TR23's 1,224 ISO weeks. Find the week's mean, median, 20% trimmed mean, compound daily rate and mean log return.

SOLUTION

The sum is −7.4669%, so the mean is −1.493%. Sorted, the returns are −9.6997%, −5.7865%, −4.0161%, +2.7842% and +9.2512%, so the median is the third, −4.016%. The 20% trimmed mean drops one return at each end and averages the middle three: −2.339%.

The growth factor is $\prod_t (1+R_t) = 0.91697$, a loss of 8.303%. By (1.9) in days, $R_G = 0.91697^{1/5} - 1 = -1.719\%$, and by (1.12) the mean log return is $\bar r = \frac{1}{5}\ln 0.91697 = -1.734\%$. Every centre answers a different question.

DEFINITION **Mean, median and trimmed mean**

Let $R_{(1)} \le \cdots \le R_{(T)}$ be the sorted returns. The **sample mean** is $\bar R = \frac{1}{T}\sum_t R_t$, and the **sample median** is $\text{med} := \hat F^{-1}(\frac{1}{2}) = R_{(\lceil T/2 \rceil)}$ (LM14). For a count $0 \le j_0 < T/2$, the **trimmed mean** drops the j_0 smallest and the j_0 largest returns and averages the rest:

$$\frac{1}{T-2j_0}\sum_{j=j_0+1}^{T-j_0} R_{(j)}.$$

The **5% trimmed mean** takes $j_0 = \lfloor 0.05T \rfloor$.

A statistic is **robust** when a few observations cannot move it far. Raising the largest return by any amount moves the mean without limit, but leaves the median and a trimmed mean unchanged.

Two more centres concern growth. By (1.12), $\ln G = \sum_t r_t = T\bar r$, so the compound daily rate of (1.9) is $R_G = G^{1/T} - 1 = e^{\bar r} - 1$, slightly above $\bar r$. Under $\hat F$, (14.12) gives $R_G \le \bar R$, and by (3.9) the gap is about $s^2/2$, the volatility drag of LM3.

EXAMPLE 2

Five centres of TAIEX's day

Compute TR23's mean, median, 5% trimmed mean, compound daily rate and mean log return, and explain their order.

SOLUTION

The mean is $\bar{R} = 0.062505\%$, or 0.0625%, and the median is the 2,925th return, 0.0879% (LM14). The 5% trimmed mean drops $j_0 = \lfloor 0.05 \times 5{,}849 \rfloor = 292$ returns at each end and gives 0.0832%. For growth, $\bar{r} = 0.055170\%$, and LM14's compound rate is $R_G = e^{\bar{r}} - 1 = 0.055185\%$. Exhibit 3 lists the first three.

The median and the trimmed mean sit above the mean because large falls outnumber large rises, 108 against 67 beyond ±3% (LM15). The compound rate sits below the mean by the drag: $0.062505\% - 0.055185\% = 0.7320$ bp, against $s^2/2 = 0.7301$ bp with $s = 1.2084\%$.

PITFALL **A robust centre answers another question**

On a skewed law the median and a trimmed mean differ from the mean even without sampling noise, because they describe other points of the distribution. Growth and the Kelly fraction (LM4) depend on the mean. Use a robust centre to describe the typical day, not to replace $\bar{R}$ in a formula.

KNOWLEDGE CHECK 1

In Example 1, lower the first return by 10 points, from −9.6997% to −19.6997%. What happens to the mean, the median and the 20% trimmed mean?

ANSWER

The mean falls by 10/5 = 2 points, to −3.493%. The changed return is still the lowest, so the median (−4.016%) and the 20% trimmed mean (−2.339%) do not move.

3 HOW FAR RETURNS SPREAD

- ☐ calculate the standard deviation with divisor T and $T - 1$, the mean absolute deviation, the interquartile range and the median absolute deviation, and convert the last two into normal-equivalent volatilities

Spread is measured around a centre: by squared deviations, by absolute deviations or by quantiles. The book's standard deviation divides by T, $s^2 = \frac{1}{T}\sum_t (R_t - \bar{R})^2$, the variance of $\hat{F}$ (LM14). Software often divides by $T - 1$ instead:

$$s_{\mathrm{u}}^2 := \frac{T}{T-1}\, s^2. \tag{19.1}$$

LM20 shows that s_{u}^2 is an unbiased estimator of σ^2 for iid draws, which is why software often uses it. NumPy's `std` divides by T by default (Section 8). On large samples the two barely differ: LM10 found a factor of 1.000114 at 4,382 days.

DEFINITION **Absolute deviations and quartiles**

The **mean absolute deviation** is $\frac{1}{T}\sum_t |R_t - \bar{R}|$. The **interquartile range** is $\mathrm{IQR} := \hat{F}^{-1}(0.75) - \hat{F}^{-1}(0.25)$. The **median absolute deviation** MAD is the median of $|R_t - \mathrm{med}|$, $t = 1, \dots, T$, by the same quantile rule.

To compare these scales with s, compute them for a normal law. Let $R \sim \mathcal{N}(\mu, \sigma^2)$ and $Z = (R - \mu)/\sigma$. Because $z\varphi(z) = -\varphi'(z)$, symmetry gives $\mathbb{E}|Z| = 2\int_0^\infty z\varphi(z)\,dz = 2\varphi(0) = \sqrt{2/\pi}$. The median z of $|Z|$ solves $\Pr(|Z| \le z) = 2\Phi(z) - 1 = \frac{1}{2}$, so $z = \Phi^{-1}(0.75) = 0.6745$, and by (15.6) the quartiles of R are $\mu \pm \sigma\Phi^{-1}(0.75)$.

RESULT **Normal-equivalent scales**

For $R \sim \mathcal{N}(\mu, \sigma^2)$,

$$\begin{aligned} \mathbb{E}|R - \mu| &= \sigma\sqrt{2/\pi}, \\ F^{-1}_{|R-\mu|}(\tfrac{1}{2}) &= \sigma\,\Phi^{-1}(0.75), \\ \mathrm{IQR} &= 2\sigma\,\Phi^{-1}(0.75), \\ \sigma &= F^{-1}_{|R-\mu|}(\tfrac{1}{2}) / \Phi^{-1}(0.75), \\ \sigma &= \mathrm{IQR}\,/\big(2\Phi^{-1}(0.75)\big), \\ \sigma &\approx 1.4826\,F^{-1}_{|R-\mu|}(\tfrac{1}{2}) \approx \mathrm{IQR}\,/1.349. \end{aligned} \tag{19.2}$$

The constants $1/\Phi^{-1}(0.75) \approx 1.4826$ and $2\Phi^{-1}(0.75) \approx 1.349$ are rounded to five and four figures; the book's computations use them exactly.

On data, $1.4826\ \mathrm{MAD}$ and $\mathrm{IQR}\,/1.349$ are **normal-equivalent scales**: on a large normal sample they come out close to s.

EXAMPLE 3

The tariff week's spread

For Example 1's five returns, compute s, s_{u}, the mean absolute deviation and the MAD.

SOLUTION

From the mean, −1.4934% to four decimals, the deviations are −8.2063, −2.5227, −4.2931, +10.7446 and +4.2776 points. Their squares sum to 225.88, so $s = \sqrt{225.88/5} = 6.721\%$ and $s_{\mathrm{u}} = \sqrt{225.88/4} = 7.515\%$, 1.118 times larger. The absolute deviations average 6.009%.

From the median −4.016%, the absolute deviations are 5.684, 0, 1.770, 13.267 and 6.800 points. Their median, the MAD, is 5.684%. With five days the divisor matters.

EXAMPLE 4

A typical day against the standard deviation

Compute TR23's s, s_{u}, mean absolute deviation, IQR and MAD and the two normal-equivalent scales, and annualize the scales with $A = 246.12$.

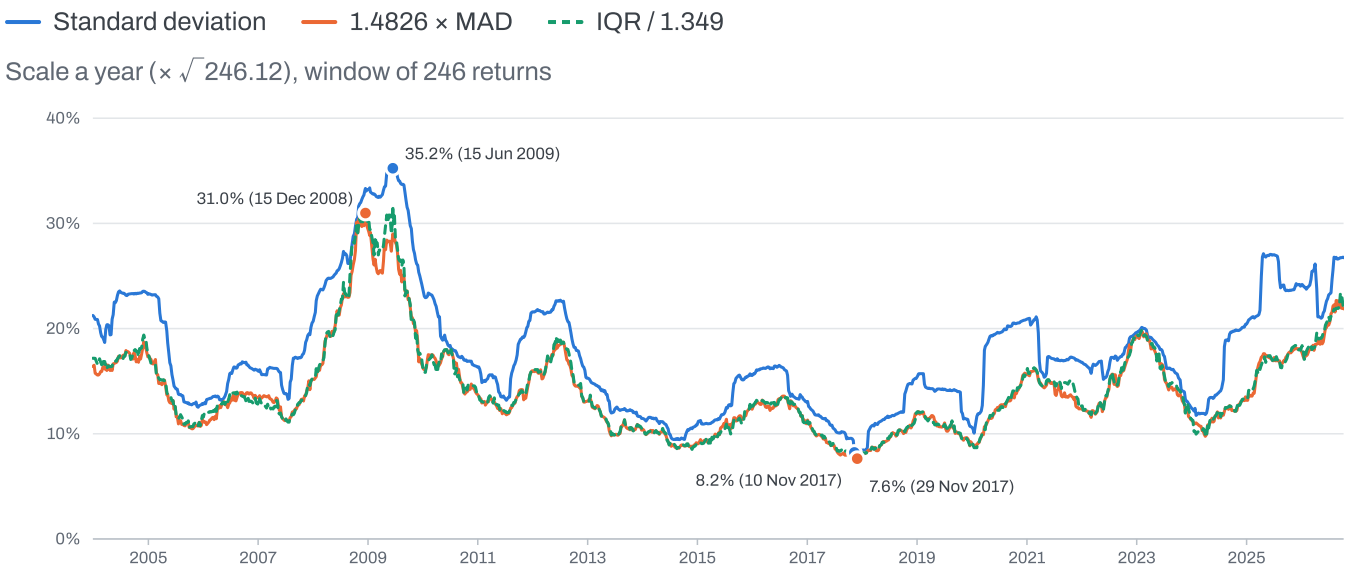
SOLUTION

TR23 has $s = 1.2084\%$ and $s_{\mathrm{u}} = 1.2085\%$. The mean absolute deviation is 0.836%, or 0.692 of s, against $\sqrt{2/\pi} = 0.798$ for a normal law. The quartiles are −0.4904% and 0.6812%, so $\mathrm{IQR} = 1.1716\%$. The MAD is 0.5889%: as no two returns lie equally far from the median, exactly $\lceil T/2 \rceil = 2{,}925$ days lie within one MAD of it.

The normal-equivalent scales are $1.4826 \times 0.5889\% = 0.873\%$ and $1.1716\%/1.349 = 0.868\%$. Annualized by the $\sqrt{A}$ convention (Exhibit 7), they give 13.7% and 13.6%, against 18.96% from s. The typical day of TAIEX moves like a 13.7% volatility, and the large days add the rest.

Exhibit 2 repeats the comparison over rolling years. The three scales rise and fall together; s stands furthest above the other two when a few extreme days fall into a calm year.

Exhibit 2: Three annual scales of TAIEX over rolling years



Note: TR23's daily simple returns in 5,604 windows of 246 consecutive returns, plotted at each window's last date, 29 December 2003 to 8 October 2026. Every scale is multiplied by $\sqrt{246.12}$, and every fifth window is plotted, plus the last and those where a scale or the ratio reaches its maximum or minimum. Window medians and quartiles are $R_{(123)}$, $R_{(62)}$ and $R_{(185)}$.

The ratio of s to $1.4826\ \mathrm{MAD}$ runs from 1.01 (window ending 17 October 2008) to 1.72 (30 March 2020). The last window gives 26.7%, 21.9% and 22.2%. Source: as Exhibit 1.

None of these scales depends on the order of the days. A drawdown does, so it is a path statistic; LM32 studies it.

> **PITFALL** **A robust scale leaves out the risk**
>
> The normal-equivalent scales treat the largest days almost as if they were absent. For a position, those days are the risk: a variance, a Kelly fraction (LM4) or a Sharpe ratio needs s. Quote a robust scale to describe the typical day.

導讀 **一般的一天與平均的一天**

敘述統計（descriptive statistics）用幾個數字描述一組報酬。平均數（mean）會被極端的日子拉動；中位數（median）與截尾平均數（trimmed mean）幾乎不受少數幾天影響，屬於穩健統計（robust statistics）。離散程度（dispersion）也一樣：標準差（standard deviation）與平均絕對偏差（mean absolute deviation）讓大漲大跌的日子占很重的分量，四分位距（interquartile range）與中位數絕對離差（median absolute deviation）只看中間那一半。換算成常態分布（normal distribution）下等效的波動率（volatility），加權股價報酬指數「一般的一天」年化（annualized）約 13.7%，標準差換算卻是 18.96%，差額來自少數大波動日。樣本變異數（sample variance）改除以 $T-1$，是為了得到不偏估計量（unbiased estimator）；只有樣本很小時，除以 T 或 $T-1$ 才有明顯差別。

KNOWLEDGE CHECK 2

By what factor does s_{u} exceed s for the five days of Example 1, and for TR23's 5,849 days?

ANSWER

By (19.1) the factor is $\sqrt{T/(T-1)}$: $\sqrt{5/4}=1.118$ for five days and $\sqrt{5{,}849/5{,}848}=1.0000855$ for TR23.

4 SKEWNESS AND KURTOSIS AS STATISTICS

- ☐ compute sample skewness and excess kurtosis, their adjusted versions and quantile-based measures, and judge which of them hang on a few days

LM14 defined skewness and kurtosis for a random variable, (14.5). Applied to $\hat{F}$ they become statistics, with a choice of divisor and a strong dependence on a few days.

DEFINITION **Sample skewness and kurtosis**

With $\bar{R}$ and s (divisor T), the **sample skewness** and **sample kurtosis** are

$$\begin{aligned}
\widehat{\mathrm{Skew}} &:= \frac{\frac{1}{T}\sum_t (R_t - \bar{R})^3}{s^3}, \\
\widehat{\mathrm{Kurt}} &:= \frac{\frac{1}{T}\sum_t (R_t - \bar{R})^4}{s^4}, \\
\widehat{\mathrm{Skew}}_{\mathrm{adj}} &:= \frac{\sqrt{T(T-1)}}{T-2}\,\widehat{\mathrm{Skew}}, \\
\widehat{\mathrm{Kurt}}_{\mathrm{adj}} &:= 3 + \frac{T-1}{(T-2)(T-3)} \\
&\quad \times \left[(T+1)(\widehat{\mathrm{Kurt}} - 3) + 6\right].
\end{aligned} \tag{19.3}$$

The sample excess kurtosis is $\widehat{\mathrm{Kurt}} - 3$; the last two are the **adjusted** versions.

Formula (19.3) is (14.5) under $\hat{F}$. It is what LM4 and LM5 computed for 0050, with $\hat{\mu} = \bar{R}$ and s, and SciPy's default (`bias=True`). SciPy returns the adjusted versions with `bias=False`; their factors tend to 1 as T grows. How far a sample skewness moves between samples would be measured by a standard error; LM20 explains why it gets none.

EXAMPLE 5

When the divisor matters

Compute the skewness and excess kurtosis, plain and adjusted, of the 23 calendar-year log returns of 2003–2025 and of TR23's daily returns. Then adjust LM4's 8.12 for 0050's 2,902 days.

SOLUTION

By (19.3) the 23 years have a skewness of −0.99 and an excess kurtosis of 2.31. Adjusted, they become −1.06 and 3.22: the skewness factor is $\sqrt{23 \times 22}/21 = 1.071$, and the excess kurtosis is $\frac{22}{21 \times 20}(24 \times 2.31 + 6) = 3.22$.

TR23's daily values move by about 0.1% at most: −0.3384 to −0.3385 and 5.268 to 5.274. LM4's 8.12 becomes 8.14 at 2,902 days.

Quantiles give shape measures that a few days cannot move far.

DEFINITION **Quartile skewness and octile kurtosis**

With quantiles $\hat{F}^{-1}(u)$ and $\mathrm{IQR} = \hat{F}^{-1}(\tfrac{3}{4}) - \hat{F}^{-1}(\tfrac{1}{4})$,

$$\begin{aligned} \mathrm{Skew}_{\mathrm{Q}} &:= \frac{\hat{F}^{-1}(\tfrac{3}{4}) + \hat{F}^{-1}(\tfrac{1}{4}) - 2\,\mathrm{med}}{\mathrm{IQR}}, \\ \mathrm{Kurt}_{\mathrm{O}} &:= \frac{1}{\mathrm{IQR}}\Big[\hat{F}^{-1}(\tfrac{7}{8}) - \hat{F}^{-1}(\tfrac{5}{8}) \\ &\qquad + \hat{F}^{-1}(\tfrac{3}{8}) - \hat{F}^{-1}(\tfrac{1}{8})\Big]. \end{aligned} \tag{19.4}$$

The quartile skewness compares the distances from the median to the two quartiles. It lies between -1 and 1 and is 0 for a symmetric law. The octile kurtosis adds the range from the fifth to the seventh octile and the range from the first to the third, and divides by the IQR. By (15.6) it is 1.233 for every normal law; it rises when the outer octiles move away from the median or the inner ones toward it.

Exhibit 3: Descriptive statistics of TAIEX daily returns

Statistic	TR23	7% limit	10% limit	Normal	TR23 + one −10% day
Returns	5,849	3,077	2,772	—	5,850
Mean (%)	0.0625	0.0464	0.0804	0.0625	0.0608
Median (%)	0.0879	0.0771	0.1055	0.0625	0.0875
5% trimmed mean (%)	0.0832	0.0696	0.0959	0.0625	0.0828
s (%)	1.208	1.257	1.152	1.208	1.215
s_{u} (%)	1.208	1.257	1.152	—	1.216
Mean absolute deviation (%)	0.836	0.878	0.789	0.964	0.838
1.4826 MAD (%)	0.873	0.899	0.833	1.208	0.873
IQR/1.349 (%)	0.868	0.896	0.834	1.208	0.869
Skewness	−0.3384	−0.2768	−0.4163	0	−0.4253
Adjusted skewness	−0.3385	−0.2770	−0.4166	—	−0.4254
Quartile skewness	0.013	0.014	0.002	0	0.013
Excess kurtosis	5.268	3.393	8.072	0	5.878
Adjusted excess kurtosis	5.274	3.400	8.089	—	5.884
Octile kurtosis	1.471	1.534	1.420	1.233	1.470

Note: daily simple returns. The limit regimes, a stability check, are the returns before 1 June 2015 and from it (s and excess kurtosis also in LM15). The normal column is the law $\mathcal{N}(\bar{R}, s^2)$ with TR23's own mean and s, in-sample; "—" marks a sample-only statistic. The last column is hypothetical. The robust scales use the exact constants of (19.2).

TR23's median and excess kurtosis are LM14's. No standard errors are given: LM20 gives those of $\bar{R}$ and s, and explains why the skewness and kurtosis get none. Source: as Exhibit 1.

The regime columns of Exhibit 3 are a stability check. From the 7% limit to the 10% limit, s fell from 1.257% to 1.152% and the excess kurtosis rose from 3.39 to 8.07 (LM15). Yet the octile kurtosis fell from 1.534 to 1.420. On 7 April 2025 alone, TR23's running excess kurtosis rose from 4.25 to 4.94 (LM15).

EXAMPLE 6

One more limit-down day

Append a hypothetical −10% day to TR23's 5,849 returns. How far does each statistic of Exhibit 3 move?

SOLUTION

See Exhibit 3's last column. Relative to the unrounded values, the mean falls 2.8%, the median and the trimmed mean 0.4%, and s rises 0.6%. The two robust scales move by less than 0.1%.

The skewness goes from −0.338 to −0.425 (−26%) and the excess kurtosis from 5.27 to 5.88 (+12%). The quartile skewness moves 1.2% and the octile kurtosis 0.05%. One day out of 5,850 moves the moment measures by 26% and 12%.

PITFALL **Kurtosis measures the far tail, not the peak**

A large excess kurtosis is often read as a peaked distribution. Yet the 10% regime has the larger excess kurtosis, 8.07 against 3.39, and the smaller octile kurtosis. Excess kurtosis averages fourth powers: the ten largest of TR23's 5,849 terms supply 35.0% of their sum (LM15). Quote it with its period, and beside a quantile measure.

導讀 **偏態與峰態是很脆弱的統計量**

樣本偏態（sample skewness）與樣本峰態（sample kurtosis）把離差（deviation）取三次方、四次方（third and fourth powers）再平均，所以由最極端的幾天決定。只要多加一個假設的 −10% 交易日，樣本偏態就從 −0.338 變成 −0.425，超額峰態（excess kurtosis）從 5.27 升到 5.88。用四分位數（quartile）與八分位數（octile）算出的四分位數偏態（quartile skewness）與八分位數峰態（octile kurtosis）只看分布（distribution）的中段與肩部（shoulders），同一天只讓它們變動 1.2% 與 0.05%。兩類統計量（statistic）回答不同的問題：2015 年 6 月漲跌幅放寬後，超額峰態從 3.39 升到 8.07，八分位數峰態卻下降。除以 T 還是用調整後（adjusted）的公式，在 5,849 天裡只差約 0.1%，在 23 個年報酬上就差很多。

KNOWLEDGE CHECK 3

TR23's quartile skewness is +0.013, while its sample skewness is −0.338. Why can the signs differ? And why can the octile kurtosis fall from 1.534 to 1.420 between the regimes while the excess kurtosis rises from 3.39 to 8.07?

ANSWER

The quartile skewness compares the middle half, where the upper quartile lies slightly further from the median than the lower one. The sample skewness is decided by the largest moves, more of which are falls. Likewise the octile kurtosis weighs the shoulders and the excess kurtosis the far tail. After June 2015 the outer octiles moved closer to the quartiles, while a few days reached further.

5 AUTOCORRELATION

- ☐ compute the sample autocorrelations of returns, absolute returns and squared returns under the book's convention, and read them against the iid band

Autocorrelation measures how a series relates to its own past, the order of the days that Sections 2 to 4 ignored.

EXAMPLE 7

Two ways to correlate a series with its past

For Example 1's five simple returns, compute the lag-1 autocorrelation by (19.5) below, and as the correlation of the four pairs (R_{t-1}, R_t). Then compare the two on TR23's log returns.

SOLUTION

With the deviations of Example 3, (19.5) divides the sum of the four products of neighbours, 31.37, by the sum of all five squares, 225.88: $\hat{\rho}_1 = 0.139$. The four pairs, each side with its own mean and standard deviation, correlate 0.237.

On TR23's 5,849 log returns the two agree: 0.0393 against 0.0394. For $|r_t|$ at lag 20 they give 0.1224 against 0.1228, the 0.123 of LM17. statsmodels' `acf` agrees with (19.5) to within 10^{-15}.

A series is **weakly stationary** when its mean, its variance and each lag covariance stay the same at every date; LM25 defines this and tests for it. Its **autocovariance** at lag k is $\gamma_k := \mathrm{Cov}(x_t, x_{t-k})$, and its **autocorrelation** is $\rho_k := \gamma_k / \gamma_0$, with $\gamma_0 = \mathrm{Var}\, x_t$.

DEFINITION **Sample autocorrelation**

For a series $x_1, \dots, x_T$ with mean $\bar{x}$ and a lag $0 \le k < T$,

$$\hat{\gamma}_k := \frac{1}{T} \sum_{t=k+1}^{T} (x_t - \bar{x})(x_{t-k} - \bar{x}), \qquad \hat{\rho}_k := \frac{\hat{\gamma}_k}{\hat{\gamma}_0}. \tag{19.5}$$

The book's convention uses the full-sample mean in both factors and divides by T at every lag, as statsmodels' `acf` does. Dividing by T rather than $T - k$ shrinks $\hat{\rho}_k$ by a factor of about $(T - k)/T$, which matters only at long lags or in short series.

RESULT **The iid band**

Let $x_1, \dots, x_T$ be iid with a finite variance, and T large. Then for each lag $k \ge 1$, $\hat{\rho}_k$ is approximately normal with mean near 0 and

$$\operatorname{Var} \hat{\rho}_k \approx \frac{1}{T}, \tag{19.6}$$

so about 95% of such autocorrelations lie within the **iid band** $\pm 1.96/\sqrt{T}$. The Deep Dive derives the variance. The approximate normality is stated without proof (Brockwell and Davis 1991, Theorem 7.2.2 and Example 7.2.1); it needs a central limit theorem for dependent terms, beyond LM17's (17.5).

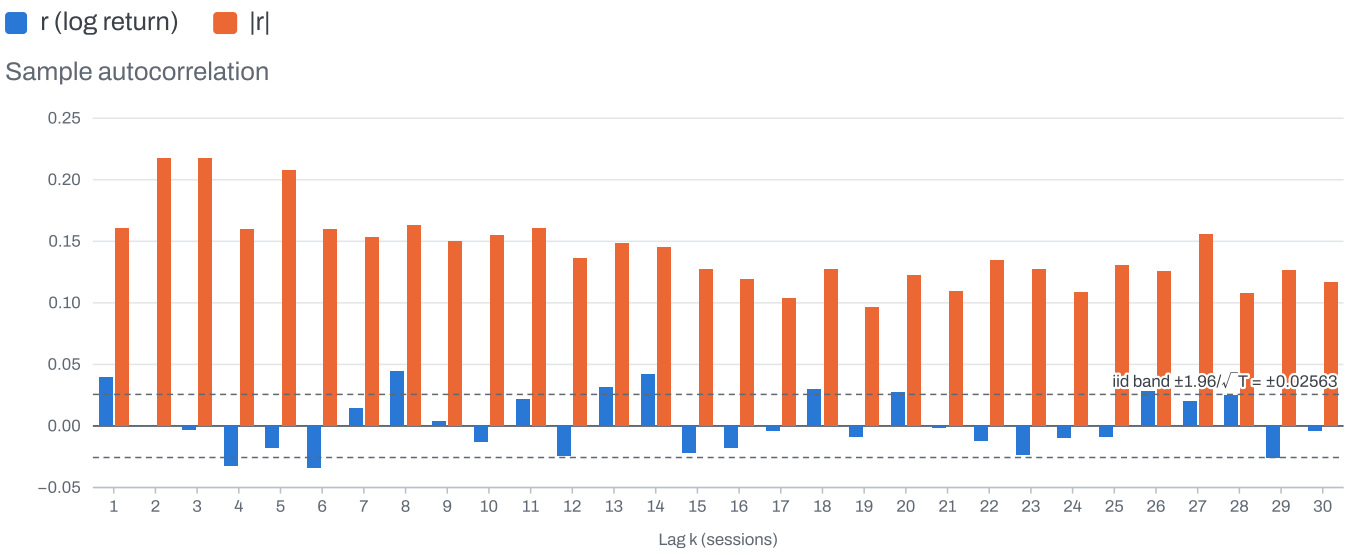
DEEP DIVE **Why the iid band is $\pm 1.96/\sqrt{T}$** optional · click to expand

Write $\tilde{x}_t := x_t - \mathbb{E}x_t$, and replace $\bar{x}$ by $\mathbb{E}x_t$ in (19.5); the error is of order $1/T$ and is dropped. The $T - k$ products $\tilde{x}_t \tilde{x}_{t-k}$, $t = k+1, \dots, T$, have mean 0 by (16.1). Two of them with different t are uncorrelated, because the latest date in their product appears once and has mean 0. Each has variance $\mathbb{E}[\tilde{x}_t^2 \tilde{x}_{t-k}^2] = \gamma_0^2$, again by (16.1). So by (16.3), $\operatorname{Var}(T\hat{\gamma}_k) \approx (T - k)\gamma_0^2$. For large T, $\hat{\gamma}_0 \approx \gamma_0$, and $\operatorname{Var} \hat{\rho}_k \approx (T - k)/T^2 \approx 1/T$.

When the squares correlate, as when large moves follow large moves, the $\tilde{x}_t$ are dependent and $\mathbb{E}[\tilde{x}_t^2 \tilde{x}_{t-k}^2]$ exceeds γ_0^2. The band is then too narrow (LM25).

On TR23 the band is $\pm 1.96/\sqrt{5{,}849} = \pm 0.0256$. In Exhibit 4, returns barely correlate with their past, but absolute returns correlate with theirs at every lag up to 30: large moves follow large moves, of either sign. This is **volatility clustering**, which LM26 models. Clustering also widens the band, because the products of the Deep Dive then vary more (LM25).

Exhibit 4: Sample autocorrelations of TAIEX log returns and absolute log returns



Note: TR23, 5,849 daily log returns, (19.5), lags 1 to 30; the band $\pm 1.96/\sqrt{T} = \pm 0.02563$ assumes iid days. For r_t, $\hat{\rho}_1 = 0.0393$, and 9 of 30 lags lie outside the band: 1, 4, 6, 8, 13, 14, 18, 20 and 26. Lag 29 (−0.02561) lies just inside it.

For $|r_t|$, $\hat{\rho}_1 = 0.161$, and the lags run from 0.096 (lag 19) to 0.217 (lag 3), all outside. For r_t^2 (not drawn), lags 1 to 3 give 0.194, 0.213 and 0.271, and no lag up to 30 falls below 0.046. For r_t^2 the band would also need a finite fourth moment of r_t, which LM15's maximum-likelihood fit ($\nu = 3.065$) lacks, so these values are description only. Source: as Exhibit 1.

The limit regimes, each about its own mean, show the same pattern. Before June 2015, $\hat{\rho}_1$ was 0.055 for r_t and 0.113 for $|r_t|$; from June 2015, 0.018 and 0.217. Squares hang on a few days: the product of 7 and 10 April 2025 alone supplies 31.5% of the lag-3 sum of products for r_t^2, against 7.7% for $|r_t|$.

> **PITFALL** **Thirty looks at one band**
>
> Under the iid model each lag falls outside the 95% band with probability about 5%. Thirty looks thus find about 1.5 bars outside by chance alone; LM21 counts such looks. TR23's nine of thirty for r_t exceed 1.5, but the band assumes iid days, and clustering widens it. Read the bars as description, not as tests.

導讀 **報酬幾乎不相關，波動卻會群聚**

自我相關（autocorrelation）衡量一個時間序列（time series）和自己過去的值有多相關，把各個落期（lag）的值排在一起就是自我相關函數（autocorrelation function）。本書的樣本自我相關（sample autocorrelation）用全樣本平均數（mean），分子（numerator）與分母（denominator）都除以 T，和 statsmodels 的 `acf` 相同。如果每天的報酬是獨立同分布（iid），約 95% 的樣本自我相關會落在獨立同分布區間（iid band）$\pm 1.96/\sqrt{T}$ 之內。加權股價報酬指數的日對數報酬（log return）自我相關很小，落後 1 期只有 0.0393；對數報酬絕對值（absolute log return）的自我相關在落後 1 到 30 期卻都在 0.096 以上，大波動之後常接著大波動，這就是波動群聚現象（volatility clustering）。群聚也讓獨立同分布區間太窄，LM25 會給出更寬的區間。

KNOWLEDGE CHECK 4

Compute $\hat{\rho}_1$ of the six values 1, −1, 1, −1, 1, −1 by (19.5). Why is it not −1?

ANSWER

The mean is 0, so $\hat{\gamma}_0 = 6/6 = 1$, and the five products of neighbours each equal −1: $\hat{\rho}_1 = -5/6 = -0.833$. Formula (19.5) divides the five products by $T = 6$, so the value shrinks by $(T-k)/T$.

6 HOW DAILY VARIANCES ADD UP

- ☐ derive the variance of a sum of m days from autocorrelations, estimate the variance ratio from overlapping and non-overlapping sums, and state the overlapping estimator's bias

Take two days of a weakly stationary series of log returns. By (16.3), $\operatorname{Var}(r_1 + r_2) = 2\gamma_0 + 2\gamma_1$, so the two-day variance is $1 + \rho_1$ times the sum of the daily variances. Independent days give 1, positively correlated days more.

RESULT **Variance of a sum of m days**

Let r_t be weakly stationary with autocovariances γ_k, and $r_t^{(m)} := r_{t-m+1} + \cdots + r_t$. Then

$$\begin{aligned} \operatorname{Var} r_t^{(m)} &= m\gamma_0 + 2\sum_{k=1}^{m-1}(m-k)\gamma_k, \\ \frac{\operatorname{Var} r_t^{(m)}}{m\gamma_0} &= 1 + 2\sum_{k=1}^{m-1}\Big(1-\frac{k}{m}\Big)\rho_k. \end{aligned} \tag{19.7}$$

Proof. By (16.3) with all weights 1, the variance is the sum of the m^2 covariances $\operatorname{Cov}(r_i, r_j)$ within the window. The m diagonal terms give $m\gamma_0$. Lag k occurs in $m-k$ pairs with $i > j$ and in as many with $i < j$, which gives $2(m-k)\gamma_k$. Dividing by $m\gamma_0$ gives the ratio. ■

The ratio in (19.7) is the model's. For uncorrelated days it is 1, and then $\sqrt{m}$ scales a volatility. As m grows, γ_0 times the model's ratio tends to the long-run variance that LM23's HAC errors estimate.

EXAMPLE 8

Two days, and LM16's year

Evaluate (19.7) at $m = 2$ on TR23, against the overlapping estimate of (19.8) below. Then compare LM16's model year, 245 days with adjacent correlation 0.04 and no other, with TR23's 246 days.

SOLUTION

At $m = 2$, TR23 gives $1 + \hat\rho_1 = 1.0393$, and the overlapping estimate is $\mathrm{VR}(2) = 1.0392$. LM16's model year gives $1 + 2(0.04)(244/245) = 1.0797$, as printed there for U28's TAIEX price index.

On TR23, with $\hat\rho_1 = 0.039337$, lag 1 alone gives $1 + 2(0.039337)(245/246) = 1.0784$. All 245 lags give 1.280 (Exhibit 6's identity column), and $\mathrm{VR}(246) = 1.208$. The other lags matter more than the first.

On data, every window of m consecutive log returns gives one sum, and the spread of these sums about $m\bar r$ is compared with m daily variances.

DEFINITION **Variance ratio**

For log returns $r_1, \ldots, r_T$ with mean $\bar r$, s_r^2 with divisor T, and $1 \le m \le T$, the **variance ratio** is

$$\mathrm{VR}(m) := \frac{\frac{1}{T-m+1}\sum_{t=m}^{T}\big(r_t^{(m)} - m\bar r\big)^2}{m\, s_r^2}. \tag{19.8}$$

The **non-overlapping** $\mathrm{VR}_{\mathrm{n}}(m)$ is the same ratio over the $\lfloor (T-o)/m \rfloor$ consecutive blocks after the first o returns, $0 \le o < m$, divided by the number of blocks.

Under iid days, $\mathrm{VR}(m)$ averages about $(1 - m/T)/(1 - 1/T)$, its **iid factor**. The estimator is biased down, by about m/T.

DEEP DIVE **The overlapping estimator under iid days** optional · click to expand

Let the r_t be iid with variance σ_r^2. A window sum minus $m\bar{r}$ is $\sum_{\text{window}} r_t - \frac{m}{T}\sum_{\text{all}} r_t$, with mean 0. The two sums share the window's m days, so by (16.3) and (17.1) its variance is

$$m\sigma_r^2 + \frac{m^2}{T^2}\,T\sigma_r^2 - 2\,\frac{m}{T}\,m\sigma_r^2 = m\sigma_r^2\left(1 - \frac{m}{T}\right).$$

Likewise $\mathbb{E}s_r^2 = \sigma_r^2(1 - 1/T)$, the bias that (19.1) removes (LM20). The ratio of these expectations is the iid factor, an approximation to the expectation of the ratio.

Experiment 1 checks it with 10,000 samples of 5,849 iid standard normal log returns from `default_rng([1, 20261011])`. The mean values of $\mathrm{VR}(m)$ are 0.960 ± 0.002 at $m = 246$ and 0.791 ± 0.005 at $m = 1{,}231$, against factors of 0.958 and 0.790. Each ± is one standard error of a mean (divisor 10,000).

Overlapping windows are far from independent. Under iid days, two consecutive one-year sums share 245 days and correlate $245/246 = 0.996$. TR23's 5,604 one-year windows hold only about $T/m = 23.8$ disjoint years, and its 4,619 five-year windows only 4.75.

Exhibit 5: Variance ratios of TAIEX log returns, 2 to 1,231 days



Note: log returns; $\mathrm{VR}(m)$ of (19.8), overlapping sums centred at $m\bar{r}$, on a logarithmic axis of m. TR23 peaks at 1.267 ($m = 127$), first falls below 1 at $m = 340$, reads 0.783 at 492 and bottoms at 0.508 ($m = 980$). 2003–2025 reads 1.054 at 246, 0.692 at 492 and 0.468 at 1,231. The dashed curve, $(1 - m/T)/(1 - 1/T)$ with $T = 5{,}849$, is what iid days would give on average. Source: as Exhibit 1.

In Exhibit 5, TR23's sums of up to 339 days vary more than the sum of their daily variances, and longer sums vary less. Exhibit 6 sets six horizons side by side.

Exhibit 6: Variance ratios at six horizons

m	Windows	T/m	$\mathrm{VR}(m)$	Identity	$\mathrm{VR}_{\mathrm{n}}, o=0$	VR_{n} range	iid factor
2	5,848	2,924.50	1.039	1.039	1.055	1.024–1.055	1.000
5	5,845	1,169.80	1.047	1.048	1.095	0.963–1.120	0.999
20	5,830	292.45	1.036	1.041	1.213	0.917–1.220	0.997
60	5,790	97.48	1.129	1.127	1.134	0.929–1.405	0.990
246	5,604	23.78	1.208	1.280	1.140	0.764–1.763	0.958
1,231	4,619	4.75	0.619	0.951	0.486	0.311–1.471	0.790

Note: TR23 log returns. Windows are the $T-m+1$ overlapping sums. Identity is (19.7) with the sample $\hat{\rho}_k$; $\mathrm{VR}(m)$ stays close to it up to $m=60$ and departs from it at 246 and 1,231 days. The non-overlapping blocks are centred at $m\bar{r}$ of the whole sample, from the first return ($o=0$) and lowest to highest over $o=0,\ldots,m-1$. The iid factor is (19.8)'s. Source: as Exhibit 1.

EXAMPLE 9

LM18's five-year windows

LM18's 4,619 windows of 1,231 sessions have a standard deviation of $\ln G$ of 0.283 about their own mean, 0.502, against $\sqrt{m}\,s_r = 0.425$ for independent days. What does (19.8) say, and how much can these windows show?

SOLUTION

Formula (19.8) centres the sums at $m\bar{r} = 0.679$, not at the windows' own mean, and gives $\mathrm{VR}(1{,}231) = 0.619$. Under iid days it would average only 0.790, far below 1, because the windows hold just 4.75 disjoint periods.

The ratio $0.619/0.790$, about 0.78, says that on TR23 the spread grows more slowly than $\sqrt{m}$. But it rests on fewer than five periods; LM22 puts an interval on the windows' spread.

PITFALL **Overlapping windows are not independent years**

A long-horizon variance ratio rests on few disjoint periods: 23.8 for TR23's one-year windows, 4.75 for its five-year ones. Under iid days the overlapping estimator is also biased down by about m/T. LM25 turns $\mathrm{VR}(m)$ into a test, and LM22 puts an interval around it.

KNOWLEDGE CHECK 5

In a model with $\rho_1 = 0.04$ and $\rho_2 = 0$, by what factor does (19.7) multiply $3\gamma_0$ for a sum of three days?

ANSWER

Lag 1 has weight $1 - \frac{1}{3} = \frac{2}{3}$ and lag 2 adds nothing, so the ratio is $1 + 2 \cdot \frac{2}{3} \cdot 0.04 = 1.0533$.

7 ANNUALIZING, AND THE ANNUAL VOLATILITY

- ☐ apply the annualization rules for the mean, variance, volatility and Sharpe ratio, state the condition each needs, and explain why TAIEX's annual volatility depends on how it is measured
- ☐ build weekly, monthly and yearly returns from daily data, and compute every statistic of the module in Python with explicit divisors and quantile rules

Each annualization rule has a condition. A year's log return is the sum of its daily log returns, (1.12), so by (14.3) the annual mean log return is $A\bar{r}$, exactly. Variances add only when days are uncorrelated, (19.7): then the annual variance is As_r^2 and the volatility $\sqrt{A}\, s_r$. A Sharpe ratio with $r_f = 0$, mean over standard deviation, then scales by $A/\sqrt{A} = \sqrt{A}$.

With correlated days, (19.7) changes the denominator. For sums of log returns with a constant mean, under (19.7)'s conditions, the Sharpe ratio of sums of m days is

$$\mathrm{SR}(m) = \frac{m\,\mathbb{E}r_t}{\sqrt{\mathrm{Var}\, r_t^{(m)}}} = \frac{\sqrt{m}\,\mathrm{SR}}{\sqrt{\mathrm{Var}\, r_t^{(m)}/(m\gamma_0)}}, \tag{19.9}$$

estimated with $\mathrm{VR}(m)$ in the denominator. Lo (2002) shows that the $\sqrt{12}$ rule for monthly Sharpe ratios holds only under special conditions. Serial correlation, he finds, can overstate an annual Sharpe ratio by as much as 65%. LM33 gives the Sharpe ratio's standard error when returns are correlated.

Weekly, monthly and yearly log returns are sums of daily log returns over calendar periods, as in Exhibit 1. Exhibit 7 applies the four rules to TR23. On simple returns these rules are conventions. Even for iid draws from $\hat{F}$, a year of A days has a simple return with mean 16.62%, standard deviation 22.29% and Sharpe ratio 0.746.

Exhibit 7: Annualization rules on TR23

Statistic	Per day	Rule	A year	Exact when
Mean simple return	0.0625%	$\times A$	15.38%	never (a convention), and not a growth rate
Mean log return	0.05517%	$\times A$	13.58%	always: sums, (1.12) and (14.3)
Variance	s^2 (s_r^2)	$\times A$	0.0359 (0.0361)	log returns on uncorrelated days, (19.7); never on simple returns
Volatility	1.2084% (1.2107%)	$\times\sqrt{A}$	18.96% (18.99%)	log returns on uncorrelated days, (19.7); never on simple returns
Sharpe ratio, $r_f = 0$	0.0517 (0.0456)	$\times\sqrt{A}$	0.81 (0.715)	log returns on uncorrelated days with a constant mean, (19.9); never on simple returns

Note: TR23, $A = 246.12$; simple returns first, log returns in brackets. By (19.9) with $\mathrm{VR}(246) = 1.208$, the log-return Sharpe ratio becomes 0.650 a year. Annual values are computed from unrounded daily ones. The figures 15.38%, 18.96%, 0.0517 and 0.81 are the volume's: LM20 gives their standard errors. Source: as Exhibit 1.

EXAMPLE 10

TAIEX's annual volatility, resolved

On 2003–2025, compare the annual volatility from independent days, from overlapping one-year sums and from the calendar years. Then cut the same 23 years into one-year blocks that start on other days.

SOLUTION

Independent days give $\sqrt{246}\, s_r = 18.563\%$. The overlapping one-year sums give $\sqrt{1.0536} \times 18.563\% = 19.05\%$, and 19.48% after dividing $\mathrm{VR}(246)$ by its iid factor, 0.957. The calendar years give 22.207%, which would need a ratio of 1.43. Autocorrelation explains only a small part of the gap.

Now skip the first o returns, $o = 0, \ldots, 245$, and cut the rest into consecutive blocks of 246 days: 23 blocks for $o \le 6$, 22 after. The 246 grids give standard deviations from 11.89% to 25.04%, with median 19.13% (Exhibit 8).

Of the 246 grids, 46 reach the calendar years' 22.207%, and 113 stay at or below 18.563%. The calendar years, of 239 to 251 sessions, are one more such cut, so most of the gap reflects how few years there are; LM20 measures that sampling error.

Exhibit 8: One-year standard deviations of TAIEX from 246 grids, 2003–2025



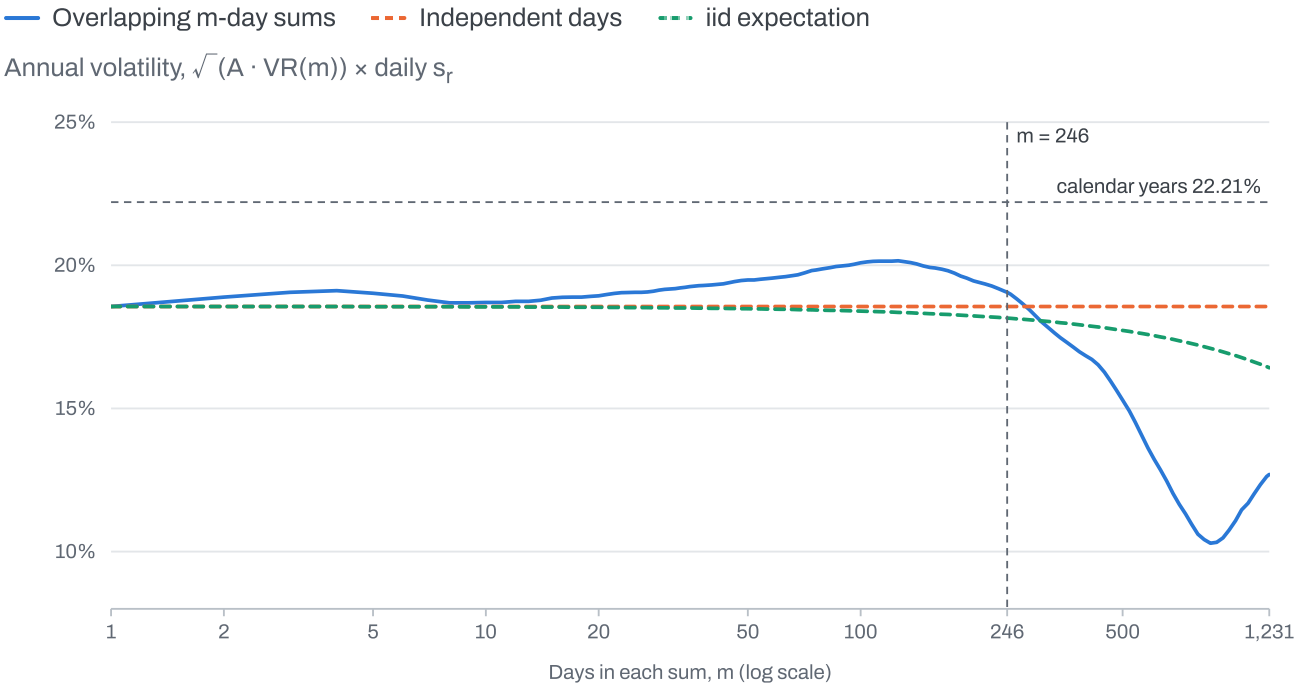
Note: 2003–2025 log returns cut into consecutive blocks of 246 sessions after skipping $o = 0, \ldots, 245$ returns (23 blocks for $o \leq 6$, 22 after); a year is 246 sessions (LM17). Each standard deviation is about the blocks' own mean, with the number of blocks as divisor. Median 19.13%, the 123rd of the 246 sorted values; 18.7% of the grids lie at or above 22.207%, and 45.9% at or below 18.563%. The spread is descriptive, not an interval (LM20). Source: as Exhibit 1.

Adding 2026's 185 sessions changes the long horizons. TR23's $\mathrm{VR}(246)$ is 1.208, against 1.054 on 2003–2025: the index's 73.0% rise in 2026's nine months (LM17) moves a long-horizon ratio by itself.

The widget redraws Exhibit 5 as an annual volatility, $\sqrt{A\,\mathrm{VR}(m)}\,s_r$, for any m up to 1,231, with $A = 246$ on 2003–2025 (LM17's year) and 246.12 on TR23. It opens on 2003–2025 at $m = 246$: $\mathrm{VR}(246) = 1.0536$ and 19.05% against 18.56% for independent days. The 5,419 windows hold 23.0 disjoint periods, and the iid factor is 0.957.

How daily variances add up

INTERACTIVE



m · years (m / A)

246 · 1.00

Windows T – m + 1 · disjoint periods T / m

5,419 · 23.0

VR(m) · iid factor · their ratio

1.0536 · 0.957 · 1.101

Annual volatility: overlapping sums · independent days

19.05% · 18.56%

TAIEX total-return log returns; descriptive. VR(m) is (19.8): every window of m consecutive days, centred at m times the mean. The iid expectation is the average of VR(m) for independent days, (1 - m/T)/(1 - 1/T). Overlapping windows are not independent: the sample holds only T/m disjoint periods.

PITFALL **A mean simple return times A is not a growth rate**

TR23's $A\bar{R} = 15.38\%$ is the sum of 246.12 average days, not a growth rate. The index's annual log growth is $A\bar{r} = 13.58\%$, and its 23 calendar years averaged 14.97% in simple returns (LM14). Use $A\bar{r}$ for growth, and name the mean behind any annual figure.

導讀 **年化：每一條規則都有條件**

對數報酬（log return）可以相加，所以日平均數（mean）乘以 A 就是年平均數，這條年化（annualization）規則永遠成立。對數報酬的變異數（variance）乘以 A、波動率（volatility）與夏普比率（Sharpe ratio）乘以 $\sqrt{A}$，都要求序列平穩（stationary）而且每天的報酬互不相關（uncorrelated）；簡單報酬（simple return）套用這些規則只是慣例。有自我相關（autocorrelation）時，要用變異數比（variance ratio）修正。加權股價報酬指數 2003–2025 年用日資料換算的年波動率是 18.56%，用重疊的一年期報酬是 19.05%，23 個日曆年卻是 22.21%。同樣的 23 年從不同的日子切成一年一段，標準差（standard deviation）從 11.89% 到 25.04% 都有，所以差距主要來自年數太少，而不是日變異數加不起來。這些都只是描述；23 個值的抽樣誤差（sampling error）要用標準誤（standard error）衡量，見 LM20。

KNOWLEDGE CHECK 6

TR23's daily log returns have $\bar{r}/s_r = 0.045567$, with $r_f = 0$. Annualize this Sharpe ratio for uncorrelated days with $A = 246.12$, and by (19.9) with $\mathrm{VR}(246) = 1.2080$.

ANSWER

For uncorrelated days, $0.045567 \times \sqrt{246.12} = 0.7149$, or 0.715 a year. By (19.9), $0.7149/\sqrt{1.2080} = 0.650$: positive autocorrelation over the year lowers the annual Sharpe ratio.

8 DESCRIPTIVE STATISTICS IN PYTHON

- [ ] build weekly, monthly and yearly returns from daily data, and compute every statistic of the module in Python with explicit divisors and quantile rules

Each library has its own default divisor and quantile rule, so pass them explicitly.

PYTHON

```python
import numpy as np
import polars as pl
from scipy import stats
from statsmodels.tsa.stattools import acf

tr = pl.read_csv("data/taiex_tr.csv")["price"].to_numpy()
R = tr[1:] / tr[:-1] - 1                # T = 5,849 simple returns
r = np.log1p(R)                         # log returns, for sums
s, s_u = R.std(ddof=0), R.std(ddof=1)   # always pass ddof
med = np.quantile(R, 0.5, method="inverted_cdf")      # R_(2925)
mad = np.quantile(np.abs(R - med), 0.5, method="inverted_cdf")
j0 = int(0.05 * len(R))                 # 292 dropped at each end
trim5 = np.sort(R)[j0:-j0].mean()
print(stats.skew(R), stats.kurtosis(R))  # -0.3384, 5.268
print(stats.kurtosis(R, bias=False))     # 5.274, adjusted


def rho(x, k):                          # (19.5): full mean, divisor T
    xc = x - x.mean()
    return (xc[k:] * xc[:-k]).sum() / (xc * xc).sum()


def vr(x, m):                           # (19.8), overlapping sums
    c = np.concatenate([[0.0], np.cumsum(x)])
    d = c[m:] - c[:-m] - m * x.mean()
    return (d**2).mean() / (m * x.var(ddof=0))


print(rho(r, 1), acf(r, nlags=30, fft=False)[1])   # 0.0393, 0.0393
print(vr(r, 246))                                  # 1.208
d = pl.read_csv("data/taiex_tr.csv", try_parse_dates=True)
d = d.with_columns(r=pl.col("price").log().diff()).drop_nulls()
yr = (d.filter(pl.col("date").dt.year() <= 2025)
       .group_by(pl.col("date").dt.year().alias("y"))
       .agg(pl.col("r").sum()).sort("y"))
print(yr["r"].std(ddof=0))              # 0.22207, divisor 23
```

NumPy's `std` divides by T unless given `ddof=1`, and Polars' by $T-1$ unless given `ddof=0`. SciPy's `skew` and `kurtosis` use divisor-T moments unless given `bias=False`. `np.median` averages the two middle values of an even sample, so the book's rule needs `method="inverted_cdf"`. statsmodels' `acf` already uses the book's convention.

IN PYTHON **Companion lab**

The Polars-first notebook `LM19_lab.ipynb` rebuilds the module's exhibits and computational examples from TR23's closes, computes the widget's data and solves Practice Problems 7 to 12. It ends with a stability check: the variance ratios within each limit regime.

The notebook reads FinMind files that you download once with your own token. From the repository root, run `uv run python data/fetch_finmind.py`, then `uv run python data/make_lab_data.py`. FinMind's licence does not allow the book to redistribute the data. After that, the notebook runs offline.

KNOWLEDGE CHECK 7

A colleague computes the standard deviation of the 23 calendar-year log returns with Polars' `std()` and gets 22.706%. Why does it differ from LM15's 22.207%?

ANSWER

Polars' `std` divides by $T - 1 = 22$ by default, and LM15 by 23. The factor is $\sqrt{23/22} = 1.0225$, and $22.207\% \times 1.0225 = 22.71\%$. Pass `ddof=0` for the book's convention.

SUMMARY

- Robust centres describe the typical day; growth needs the mean.
- The standard deviation divides by T or, as s_{u}, by $T - 1$, (19.1); (19.2) makes robust scales normal-equivalent.
- Sample skewness and kurtosis, (19.3), hang on a few days; the quantile measures of (19.4) do not.
- TAIEX's returns barely autocorrelate, (19.5), its absolute returns strongly; the iid band (19.6) assumes what clustering breaks.
- The variance of a sum of m days depends on every lag up to $m - 1$, (19.7); $\mathrm{VR}(m)$, (19.8), measures it, biased down by about m/T under iid days.
- Log means annualize exactly; log variances, volatilities and Sharpe ratios only for uncorrelated days, (19.9); on simple returns the rules are conventions.
- TAIEX's annual volatility depends on how it is measured, and 23 calendar years pin it down loosely.

PRACTICE PROBLEMS

1. TR23's simple returns on 16–20 March 2020 were −4.0582%, −2.8622%, −2.3407%, −5.6081% and +6.3683%. Their 20% trimmed mean, dropping one return at each end, is *closest* to:

   A. −1.700%

B. −2.862%

C. −3.087%

2. The 23 calendar-year log returns of 2003–2025 have a standard deviation of 22.207% with divisor 23. With divisor 22, s_{u} is *closest* to:

   A. 22.71%

   B. 23.22%

   C. 21.72%

3. TR23's MAD of daily simple returns is 0.5889% (quantiles $R_{(\lceil uT \rceil)}$). With the constant 1.4826 of (19.2) and $A = 246.12$, the normal-equivalent annual volatility is *closest* to:

   A. 9.24%

   B. 13.70%

   C. 6.23%

4. TR23's quartiles of daily simple returns are −0.490%, 0.0879% and 0.681% (LM14). The quartile skewness of (19.4) is *closest* to:

   A. 0.013

   B. −0.013

   C. 0.163

5. In a model with $\rho_1 = 0.0393$ and no other autocorrelation, the ratio of (19.7) at $m = 2$ is *closest* to:

   A. 1.0786

   B. 1.0195

   C. 1.0393

6. Daily log returns are iid, and $T = 5{,}849$. The average of $\mathrm{VR}(1{,}231)$ of (19.8) is *closest* to:

   A. 1.00

   B. 0.79

   C. 0.21

7. TR23's daily log returns have $s_r = 1.2107\%$ (divisor T). Take $m = 246$ days and LM16's assumption: adjacent days correlate 0.04, other pairs not at all. Find the model's ratio of (19.7), to four decimals, and the annual volatility $\sqrt{246}\, s_r$ with and without it, to two decimals. Compare TR23's lag 1 alone, 1.0784, and $\mathrm{VR}(246) = 1.208$: what do the other lags add?

8. (a) Derive (19.8)'s iid factor from (16.3) and (17.1). (b) Evaluate it at $m = 246$, $T = 5{,}664$ and at $m = 1{,}231$, $T = 5{,}849$, to three decimals. (c) Divide 2003–2025's $\mathrm{VR}(246) = 1.0536$ by its factor, to three decimals, and turn the result into an annual volatility with $\sqrt{246}\, s_r = 18.563\%$, to two decimals.

9. On 2003–2025, daily log returns have $\bar{r}/s_r = 0.0400$ ($r_f = 0$). (a) Annualize it by $\sqrt{246}$ and by (19.9) with $\mathrm{VR}(246) = 1.0536$, to three decimals. (b) The 23 calendar-year log returns have mean 11.646% and standard deviation 22.207% (divisor 23); compute their ratio, to three decimals. (c) Why is it lower, although 246 times the daily mean log return, 11.634%, nearly equals 11.646%?

10. *(Python)* Let R follow the scaled Student t of (15.11), with standard deviation σ and $\nu > 2$. (a) Show that the median of $|R - \mu|$, divided by $\Phi^{-1}(0.75)$ and by σ, equals $F_{t_\nu}^{-1}(0.75)\sqrt{(\nu-2)/\nu}\,/\,\Phi^{-1}(0.75)$, where $F_{t_\nu}^{-1}$ is the t_ν quantile function (`stats.t.ppf`). (b) Evaluate it at $\nu = 5$ and at LM15's $\nu = 3.065$, to three decimals, and compare TR23's $1.4826\ \mathrm{MAD}/s$, 0.723 from unrounded values.

11. *(Python)* Compute $\hat{\rho}_k$ of TR23's r_t, $|r_t|$ and r_t^2 for $k = 1, \ldots, 30$ by (19.5) in NumPy, and check them against statsmodels' `acf(x, nlags=30, fft=False)`. Count the lags with $|\hat{\rho}_k| \geq 1.96/\sqrt{T}$, with the guard `>= band - 1e-12`. Report lag 29's value for r_t to five decimals and its margin to the band to three significant figures.

12. *(Python)* With Polars, build the weekly (ISO weeks), monthly and calendar-year log returns of 2003–2025, sorting after `group_by`. Reproduce Exhibit 1's four annual standard deviations (divisor the number of periods). Then reproduce Exhibit 8's lowest, highest and $o = 0$ values and the median of its 246 values by $\hat{F}^{-1}(\frac{1}{2})$, all in percent to two decimals.

SOLUTIONS

1. **C** is correct. Sorted, the returns are −5.6081%, −4.0582%, −2.8622%, −2.3407% and +6.3683%. The middle three average −3.087%. A is the mean, with nothing trimmed. B is the median, which trims two returns at each end.

2. **A** is correct: $22.207\% \times \sqrt{23/22} = 22.71\%$. B multiplies by $23/22$, the factor of the variance, not of the standard deviation. C divides by the square root instead of multiplying.

3. **B** is correct: $0.5889\% \times 1.4826 \times \sqrt{246.12} = 13.70\%$. A annualizes the MAD itself, which is not a standard deviation. C divides by 1.4826 instead of multiplying.

4. **A** is correct: $(0.681 - 0.490 - 2 \times 0.0879)/(0.681 + 0.490) = 0.0152/1.171 = 0.013$. B reverses the sign, taking twice the median minus the quartiles. C leaves out the median, $(0.681 - 0.490)/1.171 = 0.163$.

5. **C** is correct. At $m = 2$ the only lag is 1, with weight $1 - \frac{1}{2}$, so the ratio is $1 + 2 \cdot \frac{1}{2} \cdot 0.0393 = 1.0393$. A forgets the weight $1 - k/m$. B, 1.0195, is the square root of the correct 1.0393: a ratio of volatilities, not of variances.

6. **B** is correct: the iid factor is $(1 - 1{,}231/5{,}849)/(1 - 1/5{,}849) = 0.79$. A assumes that the estimator is unbiased. C is m/T, the size of the bias, not the ratio.

7. The model's ratio is $1 + 2(0.04)(245/246) = 1.0797$. Without it, $\sqrt{246} \times 1.2107\% = 18.99\%$; with it, $18.99\% \times \sqrt{1.0797} = 19.73\%$. TR23's lag 1 alone gives almost the same, 1.0784. All 245 lags give 1.280

by (19.7) (Exhibit 6's identity column), so the other 244 lags add about 0.20, more than the first's 0.078; $\mathrm{VR}(246) = 1.208$ points the same way.

8. (a) As in the second Deep Dive, by (16.3) and (17.1) a window sum minus $m\bar{r}$ has variance $m\sigma_r^2(1 - m/T)$, and $\mathbb{E}s_r^2 = \sigma_r^2(1 - 1/T)$. Their ratio is $(1 - m/T)/(1 - 1/T)$, a ratio of expectations in place of the expected ratio. (b) At $m = 246$, $T = 5{,}664$ it is 0.957; at $m = 1{,}231$, $T = 5{,}849$ it is 0.790. (c) The ratio is $1.0536/0.957 = 1.101$, and $\sqrt{1.101} \times 18.563\% = 19.48\%$.

9. (a) By $\sqrt{246}$, $0.0400 \times \sqrt{246} = 0.627$; by (19.9), $0.627/\sqrt{1.0536} = 0.611$. (b) The ratio is $11.646/22.207 = 0.524$. (c) The means agree because both spread the same total log growth over the years. The calendar years' standard deviation, 22.207%, exceeds even the overlapping sums' $\sqrt{1.0536} \times 18.563\% = 19.05\%$, so their ratio is lower. Autocorrelation explains little of that excess. The same 23 years cut on other days give 11.89% to 25.04% (Exhibit 8), so most of it is the sampling variation of 23 values (LM20).

10. (a) By (15.11), $R - \mu = \sigma\sqrt{(\nu - 2)/\nu}\,Y$ with $Y \sim t_\nu$. The t_ν law is symmetric, so $\Pr(|Y| \le y) = 2F_{t_\nu}(y) - 1$, which equals $\frac{1}{2}$ at $y = F_{t_\nu}^{-1}(0.75)$. The median of $|R - \mu|$ is therefore $\sigma\sqrt{(\nu - 2)/\nu}\,F_{t_\nu}^{-1}(0.75)$; divide by $\Phi^{-1}(0.75)$ and σ. (b) At $\nu = 5$, $F_{t_5}^{-1}(0.75) = 0.7267$, and the ratio is $0.7267 \times \sqrt{0.6}/0.6745 = 0.835$. At $\nu = 3.065$, the quantile is 0.7628 and the ratio 0.667. TR23's 0.723 lies between them.

11. Of the 30 lags, 9 lie outside the band for r_t, and all 30 for $|r_t|$ and r_t^2 (Exhibit 4, whose note gives the band's condition for r_t^2). Lag 29 of r_t is −0.02561, inside the band of ±0.02563 by 2.15×10^{-5}. NumPy and statsmodels agree to within 10^{-15}.

12. The four annual standard deviations are 18.56%, 18.09%, 18.12% and 22.21% (Exhibit 1). The grids run from 11.89% ($o = 171$) to 25.04% ($o = 212$), with 20.18% at $o = 0$, and their median is 19.13%, the 123rd of 246 sorted values. `np.median` would average the 123rd and 124th, 19.15%.

GLOSSARY

Term	中文	Meaning
Descriptive statistics	敘述統計	Numbers that summarize a sample, with no claim about sampling error.
Median	中位數	$\text{med} = \hat{F}^{-1}(\frac{1}{2})$, the middle return.
Trimmed mean	截尾平均數	The mean after dropping the j_0 smallest and j_0 largest values.
Robust statistics	穩健統計	Statistics that a few observations cannot move far.
Mean absolute deviation	平均絕對偏差	The mean of $\lvert R_t - \bar{R}\rvert$.
Interquartile range	四分位距	$\hat{F}^{-1}(0.75) - \hat{F}^{-1}(0.25)$.
Median absolute deviation	中位數絕對離差	MAD, the median of $\lvert R_t - \text{med}\rvert$.
Normal-equivalent scale	常態等效尺度	1.4826 MAD or IQR $/1.349$, which a normal law makes equal to σ, (19.2).
Sample skewness, kurtosis	樣本偏態、樣本峰態	(14.5) under $\hat{F}$, (19.3); the adjusted versions are SciPy's `bias=False`.
Quartile skewness, octile kurtosis	四分位數偏態、八分位數峰態	Shape measures from quantiles, (19.4).
Weakly stationary	平穩	Mean, variance and each lag covariance the same at every date (LM25).
Autocovariance	自我共變異數	$\gamma_k = \text{Cov}(x_t, x_{t-k})$; sample version (19.5).
Autocorrelation	自我相關	$\rho_k = \gamma_k/\gamma_0$, the correlation of a series with itself k dates earlier; sample version (19.5).
Autocorrelation function	自我相關函數	The autocorrelations as a function of the lag k.
iid band	獨立同分布區間	$\pm 1.96/\sqrt{T}$, which holds about 95% of iid sample autocorrelations, (19.6).

Term	中文	Meaning
Volatility clustering	波動群聚現象	Large moves followed by large moves, of either sign.
Variance ratio	變異數比	$\mathrm{VR}(m)$, the variance of overlapping sums of m days over m daily variances, (19.8).
iid factor	獨立同分布因子	$(1 - m/T)/(1 - 1/T)$, the approximate mean of $\mathrm{VR}(m)$ under iid days.

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